Ysbyty Gwynedd

Intubation of suspected COVID-19 patients on High Dependency Unit / Critical Care.

Priorities:

1. **Plan, Prepare, Communicate.**
2. **Minimise exposure to staff and equipment.**
3. **Maintain staff protection.**

Scan QR code for link to Youtube training video:

1. Decision to Intubate on HDU / Critical care	
Referral pathway	Emergency: bleep 550 ITU on call, who then liaises with ITU consultant on call. Non emergency: direct referral to ICU Consultant on call. ICU Consultant to decide on appropriateness for intubation. Once accepted: Bleep **170** for the COVID intubating anaesthetist. If 170 unavailable (eg already busy or Out of hours) choose consultant / senior trainee (ST5 or above) / most experienced intubator. Intubating anaesthetist picks up COVID intubating tray, predrawn drugs from ICU minifridge and visors. **Bleep 243 for ODP**. Patients already in ED Isolation room – Intubate them in the isolation room (see separate SOP and training video). Scan QR code on left for link to video. Patients anywhere else (eg on ward / COVID ward) – once they are accepted by ICU consultant the ward staff should be called to transfer them to HDU / critical care for intubation. **Ward staff** should transfer the patient according to the agreed protocol (with a team of outriders). Handover of care will take place on Critical care. If the patient is on C-19 ward and in extremis (likely to arrest before reaching HDU / critical care area) then move to ED isolation room to intubate.

2. Staffing required	
Inside critical care cubicle / bay	- Senior Anaesthetist (full PPE with FFP3 mask) (170 Consultant or senior trainee). - ODP (full PPE with FFP3 mask) **bleep 243.** - Do not use obstetric anaesthetist / ODP.

In Corridor	- 1 nurse (full PPE), immediately outside the door, ready to hand over any extra equipment required. - 1 runner (FFP3 mask, apron and gloves), ready get further help / equipment + responsible for scribing.

3. Equipment to *bring* to critical care (by the anaesthetist)	
Intubation tray	Ready-made intubation trays, **stored on ICU, HDU** + 2 available in **cupboard of anteroom** of ED isolation room (made up by ICU nurses and stored on COVID trolley on Cybi): - Waters Circuit - Mac 4 blade and handle (PROACT) in sealed bag - Size 7 and 8 ETT - Tube tie - Catheter mount and closed suction kit - HME filter - $EtCO_2$ line - 20 ml syringe - Sterile gel sachet - Tube clamp - Magills forceps - Feeding NG tube - NG syringe - RSI checklist
RSI Drugs	Ready-made in disposable tray in ICU fridge: made fresh at the start of each morning shift by ICU resident (all syringes capped with red bungs): - Rocuronium 1 syringe of 100 mg (10 ml) - 2 extra rocuronium ampoules and a drawing up needle - Metaraminol 0.5 mg/ml - 20 ml syringe and 200 mg Propofol ampoule (for sedation after intubation, to be drawn up as needed) and infusion line (electrocath) - Use induction agent that is safest in **your** hands. - Ketamine dose is 1-2mg/kg depending on CVS state of patient. - 20ml syringe labelled Ketamine and drawing up needle. Ketamine to be requested from ICU / HDU CD cupboard and drawn up when needed - Ketamine 10 mg/ml 20 ml : draw up neat - *(or Ketamine 50 mg/ml 10 ml : draw up 4 ml = 200 mg, dilute with 16 ml NaCl 0.9% to 20 ml total, concentration = 10 mg/ml)*
Visors	Collect 3 heavy duty visors from ICU (non-disposable) for the intubating team.

4. Equipment which is to be present on HDU / Critical Care	
Videolaryngoscope	C-MAC (One single use blade to be brought in according to need).
Airway Ancillary trolley	Metal trolley containing : C-MAC blades (1 x size 3, 1 x size 4, 1 x D-Blade, Stylet) Size 3,4,5 facemasks Bougie Ready made iv access kits : - Pink & green venflon - iv dressing - Disposable tourniquet - Swab - Alcowipe - Prefilled flush - Octopus connector Restocked from the airway drawer in the store cupboard on HDU or ITU or call ODP (243) for restocking.
Airway trolley	Drawer A : Endotracheal tube size 6 Drawer B : Supraglottic Airway for adults (3,4,5) Drawer C : Oropharyngeal airways (2,3,4) and facemasks (3,4,5) Drawer D : Front of neck access kit. To be restocked after every use by HDU / Critical care staff.
In cubicle / bay	- Ventilator with single use breathing circuit, HME, Humidification. - Monitor including $ETCO_2$ - Wall-suction with suction tubing and yankeur. - Patient table to use as work surface.
Outside each cubicle / bay	- Clinical waste bin. - Alcohol Gel.
Other	- Ultrasound for line placement. - CVP and Art line packs.

5. Preparation prior to enter HDU / Critical care area	
PPE	- Ensure whole team are feeling well, well hydrated and empty bladders. - Wash hands. - Empty Pockets – phones and bleeps to be placed in clear plastic bag. - Don PPE in HDU / Critical care PPE area. - Intubating team to Double Glove and wear an extra apron over the gown.

5. Preparation prior to enter HDU / Critical care area	
Check all equipment	- Follow the COVID Outside Room Checklist : - **Minimise exposure to staff and equipment.** - Decide who needs to go in (Suggest 2 - Maximum 3) and what equipment you will definitely need. - Discuss escalation plan. - Check : - Intubation tray - iv access - RSI drugs and sedation drugs. - Face mask of appropriate size - C-MAC blade (size 3, 4 or D blade) Stylet if using D blade - Bougie - Waters circuit with HME filter
Other	- Talk through plan including : - escalation plan - plan for failed intubation, - cardiac arrest, etc.

6. Intubation & procedures	
Pre-intubation	- Patient should have surgical facemask on and O_2 mask while you are preparing. - If the patient is already on CPAP / NIV then consider stopping this and putting patient on facemask oxygen (Non-rebreathing mask) whilst you are preparing in order for you to set up the ventilator ready. - Alternatively stop NIV and preoxygenate with Waters circuit. A second person will need to set up the ventilator. - Avoid high flow nasal oxygen. - Set up ventilator and circuit with $ETCO_2$, catheter mount and in-line suction ready to go. - Set ventilator settings ready – SIMV. PEEP 12. RR24. TV 6ml/kg. 1:1 ratio. 100% O_2. - Check all connections are tight (Push and twist technique) - Set up propofol sedation ready. - Set up $ETCO_2$ on monitor (NB – on ventilator circuit not Waters circuit) - Position C-MAC so anaesthetist and ODP can see the screen. - COVID-19 in-room RSI checklist, to be read out loud. - When ready : Turn off Oxygen for facemask before removing from patient's face. - Preoxygenate with Waters circuit (with HME) using 2 hand technique. Turn the APL so that the exhaust faces down and away from you. - ODP to give the drugs under direction and supervision of anaesthetist. - Maintain tight seal with 2 handed technique on facemask and CPAP during apnoea. - Try to avoid Bag mask ventilation. If BMV required use small volumes, low pressure with ODP gently squeezing the bag. - ODP to turn off oxygen flowmeter for the Waters circuit / facemask as soon as laryngoscopy commences. - Inflate cuff immediately once ETT inserted. Then connect immediately to ventilator circuit and start ventilating.

6. Intubation & procedures	
Post-intubation	- Read out COVID-19 post RSI checklist. - Commence sedation. - Place NG feeding tube. - Clean up any gross contamination (Vomit and respiratory secretions). - Remove outer pair of gloves and outer apron. - Wash inner pair of gloves. - Place all airway equipment back in to the intubating tray (Waters Circuit, C-MAC blade, Bougie, facemask, etc.) Place back in the clear plastic bag and **keep in the room with the patient.** - **Keep Tube clamp immediately available (on top of ventilator)**
Invasive Lines	- Stay in PPE to place any invasive lines, etc. - Scrub and place sterile gown and sterile gloves over your PPE. - CXR only after all lines and NG tube placed.
Cleaning	- Follow decontamination guidelines as set by Infection Control Policies. - Ensure adequate decontamination of C-MAC, Intubation tray, any trolley and all monitoring, ventilator, syringe drivers, etc with Green Clinell wipes. - Single use equipment not used / no longer required : - If in a sealed bag and / or washable, decontaminate with Clinell / soap and water. - If not washable (eg in paper packet) - to stay with the patient (do not dispose of).
Doffing	- Follow the PHE guide to doffing. Use buddy system - one person at a time. - Wash the visors in warm soapy water, then rinse and dry thoroughly.